

# Web Application Development

## CSS Fundamental



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-  Understand the structure, roles, and significance of CSS in a website.
-  Understand the advantages and disadvantages of different types of CSS.
-  Know how to determine the scope of CSS effects on HTML tags.



- ❑ CSS = Cascading Style Sheets
- ❑ Used to describe how elements are displayed on a web page
- ❑ Functions similarly to a template
- ❑ Can be reused for other websites
- ❑ Allows for quick changes to properties for individual pages or the entire site (cascading)



|               | Inline style Sheet                                                                                                          | Embedding style Sheet                                                                                                                                                         | External style Sheet                                                                                                                                      |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Syntax</b> | <pre>&lt;p style="color:red;"&gt; Test&lt;/p&gt;</pre>                                                                      | <pre>&lt;head&gt;&lt;style type="text/css"&gt; .highlight{color: red;} &lt;/style&gt;&lt;/head&gt; &lt;body&gt; &lt;p class="highlight"&gt; Test&lt;/p&gt;&lt;/body&gt;</pre> | <pre>&lt;head&gt;&lt;link rel="stylesheet" href="main.css" /&gt; &lt;/head&gt; &lt;body&gt; &lt;p class="highlight"&gt; Test&lt;/p&gt;&lt;/body&gt;</pre> |
| <b>Pros</b>   | <p>Easy to manage styles for each tag in the web document.</p> <p>Highest priority.</p>                                     | <p>Easy to manage styles for each web document.</p> <p>No need to load additional pages for styles.</p>                                                                       | <p>Can set styles for multiple web documents.</p> <p>Style information is cached by the browser.</p>                                                      |
| <b>Cons</b>   | <p>Must manually declare style information in each web document and other documents.</p> <p>Difficult to update styles.</p> | <p>Must declare style information for other documents each time it is used.</p>                                                                                               | <p>Time-consuming to download *.css files, slowing down the compilation process in the browser during the first use.</p>                                  |



The order of specificity for applying styles when using different types of CSS (in decreasing order of specificity)

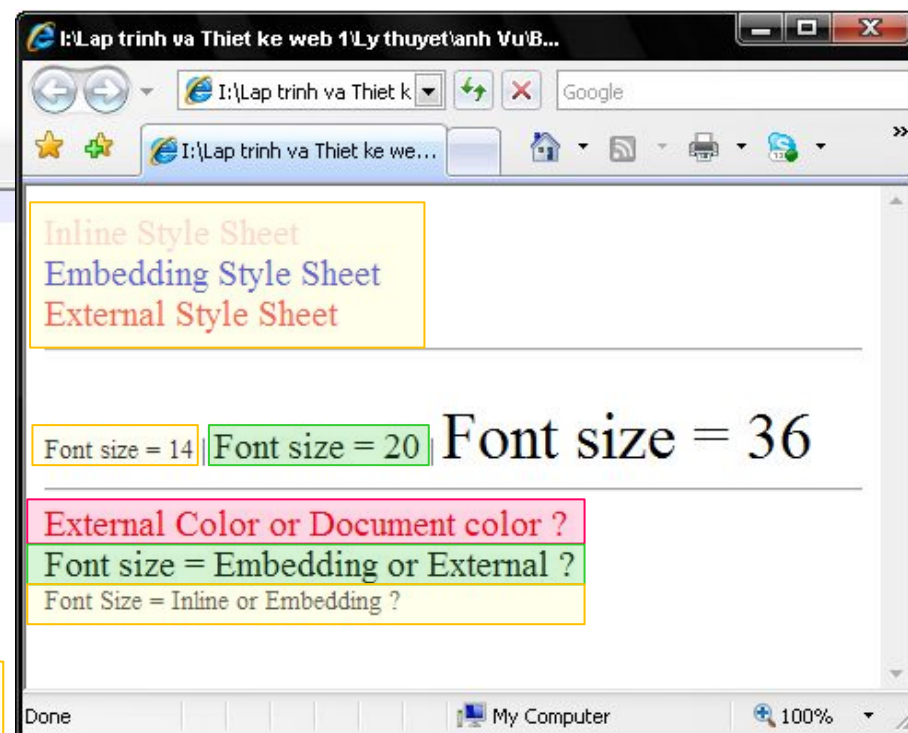
1. Inline style Sheet
2. Embedding style Sheet
3. External style Sheet
4. Browser Default
5. Inherited Element



```
PriorityCSS.css
1  MyExColor {
2    Color : Red;
3  }
4
5  div {
6    font-size:36px;
7  }
```

PriorityCSS.htm

```
1  <html>
2  <head>
3    <link rel="stylesheet" href="PriorityCSS.css" />
4
5    <style type="text/css">
6      .MyEmColor{
7        color:blue;
8      }
9      div{
10       font-size:20px;
11     }
12   </style>
13 </head>
14
15 <body>
16   <div style="color:pink"> Inline Style Sheet </div>
17   <div class="MyEmColor"> Embedding Style Sheet </div>
18   <div class="MyExColor"> External Style Sheet </div>
19   <hr>
20   <p>
21     <font style="font-size:14px">Font size = 14 </font> |
22     <font style="font-size:20px">Font size = 20 </font> |
23     <font style="font-size:36px">Font size = 36</font>
24   </p>
25   <div style="color:blue">
26     <div class="MyExColor"> External Color or Document color ? </div>
27   </div>
28
29   <div> Font size = Embedding or External ? </div>
30   <div style="font-size:14px"> Font Size = Inline or Embedding ? </div>
31 </body>
32 </html>
```





■ A CSS selector selects the HTML element(s) you want to style.

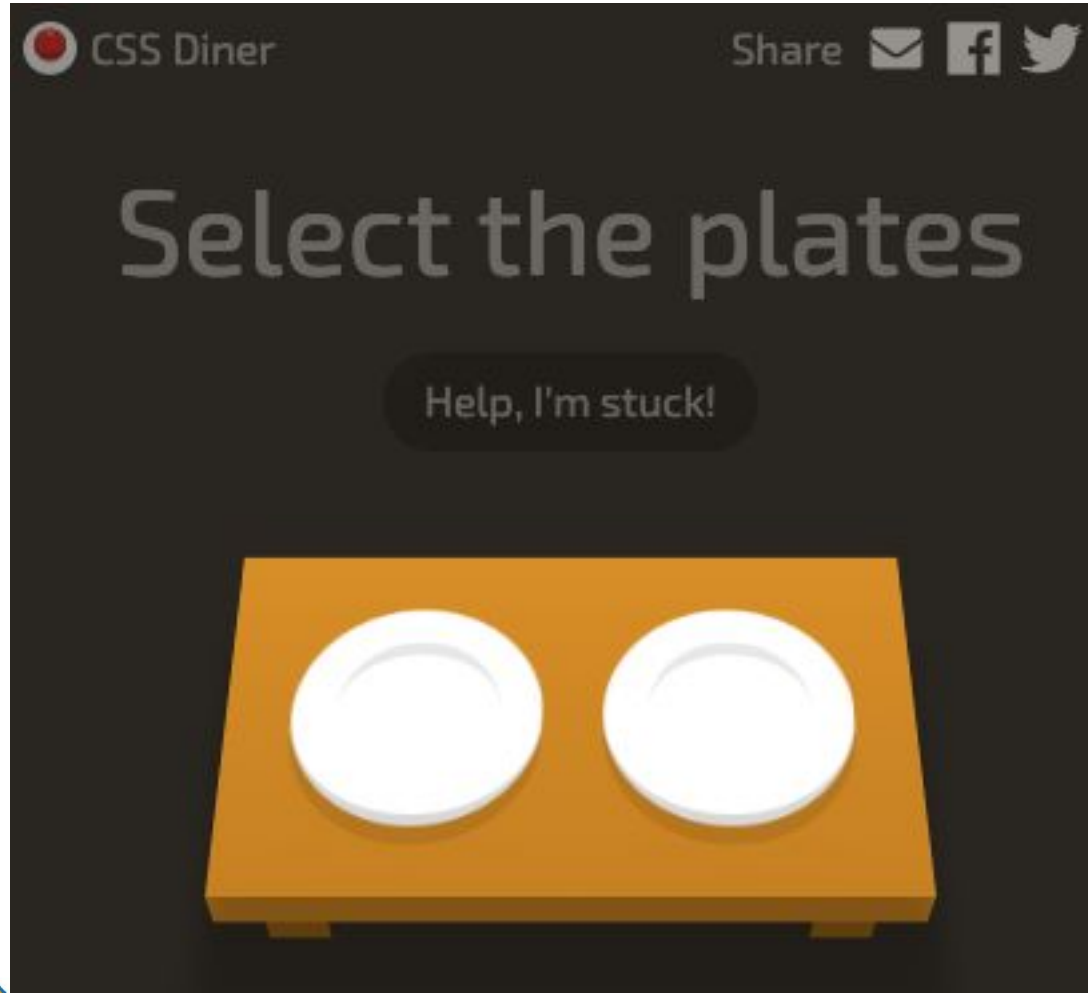
■ Selector types

- HTML element selectors
- Class selectors
- ID selectors
- ....

```
.variant1 {color: red;  
font-family: Verdana, sans-serif;  
}  
<h1 class="variant1">HCMUS</h1>
```



- ☐ Type, class, and ID selectors
- ☐ Attribute selectors
- ☐ Pseudo-classes and pseudo-elements
- ☐ Combinators





Redo the group assignment in HTML and present the page using CSS.

- ☐ Use CSS as much as possible to style the webpage (avoid using HTML tag attributes when CSS can achieve similar functionality).
- ☐ Only use an external style sheet and save it in the file **default.css**.



| Feature                                                                                                | Điểm |
|--------------------------------------------------------------------------------------------------------|------|
| Complete the functions in GA02                                                                         | 3    |
| Use external CSS to style the content of the website                                                   | 3    |
| The presentation format across web pages must be consistent                                            | 2    |
| Upload the group website content to a host (specify the URL of the website on the host in the report). | 2    |



-  <https://developer.mozilla.org/en-US/docs/Learn/CSS>
-  <http://www.w3schools.com/css/>
-  <https://flukeout.github.io>